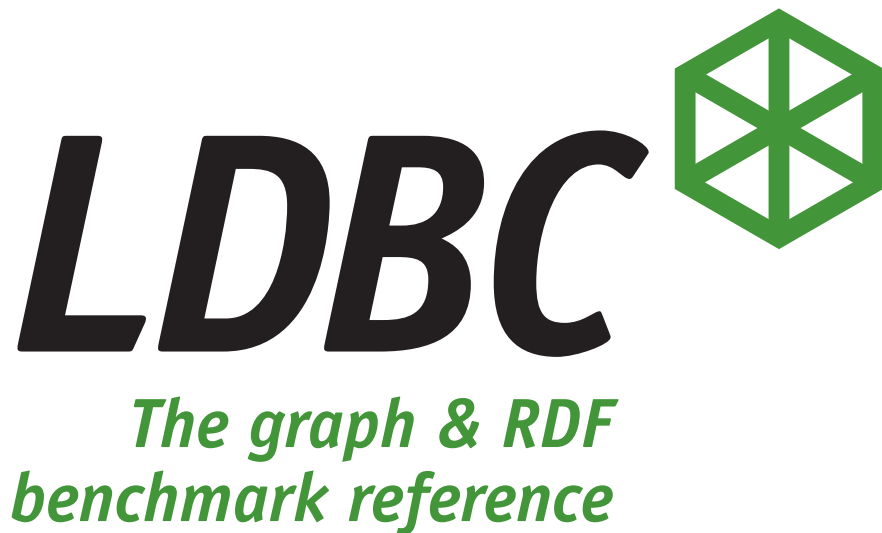




Full Disclosure Report of the LDBC Social Network Benchmark

An Implementation of the LDBC Social Network
Benchmark's Interactive Workload over Galaxybase

September 30, 2026

GENERAL TERMS

Executive Summary

The present document describes an audited run of the Interactive Workload of LDBC’s Social Network Benchmark using Galaxybase version 3.5.3. Galaxybase is a graph database developed by Zhejiang CreateLink Technology Co., Ltd. The benchmark was executed while ensuring serializable isolation level. The audited runs assessing query performance were executed on three scale factors: SF100, SF300, and SF1000. While Galaxybase has the option to run declarative Cypher queries, the benchmark queries were implemented in the Java imperative programming language. The graph schema used in the benchmark runs did not contain any precomputed attributes, and the used indices over attributes are explicitly marked in the data schema definition.

Declaration of Audit Success

This report contains an audited LDBC benchmark execution. The results have been gathered by an independent and impartial auditor who has validated the implementation of the queries, successfully run the ACID tests associated with the claimed isolation level (serializable), and verified the overall system’s configuration conformance to the description of the benchmark and its strict requirements.

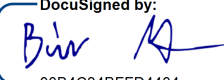
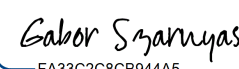
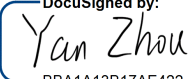
DocuSigned by:  00B4C34BFFD4404...	9/30/2026
Dr. Márton Búr	Date
(Auditor)	
DocuSigned by:  FA33C2C8CB944A5...	10/1/2026
Dr. Gábor Szárnyas	Date
(Head of LDBC SNB Task Force)	
DocuSigned by:  BBA1A13B17AE422...	10/1/2026
Yan Zhou	Date
(Test Sponsor Representative)	

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1 SYSTEM DESCRIPTION AND PRICING SUMMARY

1.1 Details of machines driving and running the workload

1.1.1 Machine overview

This benchmark a single x8aedz.metal-24xl instances, the driver and the SUT were both running on the same machine. The instance details below were obtained from the Amazon Web Services console. The operating system name and version was obtained from running `cat /etc/os-release` command.

Table 1.1: Machine Type and Location

Cloud provider	Amazon Web Services
Machine region	US-West-2
Common name of the item	x8aedz.metal-24xl
Operating system	Ubuntu 26.04 LTS

1.1.2 CPU details

The details below were obtained using the command `cat /proc/cpuinfo` (Listing A.1) and `lscpu` (Listing A.2) issued from the machine instance, and the datasheet of the used CPU type.

Table 1.2: CPU details summary

Type	AMD EPYC 9R05
Total number	2
Cores per CPU	48
Threads per CPU	48
CPU clock frequency	5 GHz
Total cache size per CPU	L1d cache: 4.5MiB L1i cache: 3MiB L2 cache: 96MiB L3 cache: 512MiB

1.1.3 Memory details

The total size of the memory installed is 3TiB, as reported by the `sudo lshw -c memory` command issued on the instance. The memory consists of 24×128GiB Samsung DIMMs, with 12 DIMMs installed per CPU. The `lshw` output identifies the memory as DIMM Synchronous Registered (Buffered) 6400 MHz, with each DIMM operating at a reported clock of 2105MHz. For the full output of the `sudo lshw -c memory` command, see Listing A.3.

1.1.4 Disk and storage details

The disk controller and motherboard type were not obtainable from the virtual machine instance. The storage consists of two individual 3.5TiB NVMe storage devices, both formatted with the XFS filesystem. In addition, the instance has a 100GB Amazon Elastic Block Store (EBS) volume formatted with the ext4 filesystem, which hosts the operating system. The storage type and capacity of the NVMe instance-storage devices are specified by Amazon Web Services for the corresponding EC2 instance type <https://aws.amazon.com/ec2/instance-types/> (accessed: September 30, 2026). Actual disk parameters for the SUT instance were obtained using `sudo lshw -c storage -c disk` and `sudo lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,MODEL,SERIAL,TRAN`, and are shown in Listing A.4 and Listing A.5.



The 4KB QD1 write performance was measured with the `fio` command and the output (Listing A.6) showed an average of 60 803 IOPS.

1.1.5 Network details

The benchmark presented ran in a client-server architecture on a single machine. For this reason, network details are not relevant and thus skipped from this document.

1.1.6 Machine pricing

The system pricing summary is included in the table below.

Table 1.3: Pricing summary

Item	Price
x8aedz.metal-24xl 3-year EC2 Instance Savings Plan (Paid All Upfront)	272 837 USD
Software license	155 200 USD
Maintenance fee (3 years)	55 872 USD

1.1.7 System availability

The latest software version of Galaxybase (version 3.5.3) was made available on April 02, 2025. This version was deployed to the machine described in this section.

2 DATASET GENERATION

2.1 General information

The data generation settings of the LDBC Datagen are described below.

Table 2.1: Datagen settings summary

Datagen version	v1.0.0
Output format	CsvComposite serializer, StringDateFormatter
Scale factors	10, 100, 300, 1000

Note: Scale factor 10 is used for query results validation only, while the other ones were used for performance measurements.

2.2 Datagen configurations

The Datagen configuration for the benchmark are not part of the audited package, as the pregenerated datasets available under the official benchmark website¹ for the respective parameters were used.

2.3 Data loading and data schema

The data is passed to the utility `AddColumn.java` to enrich the "hasCreator" edges with the post/comment's timestamp. Then, the CSVs can be loaded with `load_graph.sh` script). The loading script passes a schema descriptor in JSON format (see Listing 2.1 and Listing A.7) that defines the data types represented in the graph, as well as a mapping configuration (see Listing 2.2 and Listing A.8) that specifies which CSV columns are used to populate the respective edges and nodes in the graph. On a technical level, these JSONs are passed on to the respective load endpoint of the DB service inline by the `load_graph.sh` script.

Listing 2.1: Excerpt from schema JSON, describing the data schema

```

1 {
2   "name": "ldbc_snb",
3   "storageV2": true,
4   "storageEngine": "storage-v2",
5   "store_type": "mutable_csr",
6   "schema": {
7     "vertex_types": [
8       {
9         "type_id": 0,
10        "type_name": "PLACE",
11        "properties": [
12          {
13            "property_id": 0,
14            "property_name": "id",
15            "property_type": {
16              "primitive_type": "DT_SIGNED_INT64"
17            }
18          },
19          {
20            "property_id": 1,
21            "property_name": "name",
22            "property_type": {

```

¹<https://ldbcouncil.org/benchmarks/snb/datasets/#snb-interactive-v1>

```

23         "varchar": {
24             "max_length": 256
25         }
26     },
27 },
28 {
29     "property_id": 2,
30     "property_name": "url",
31     "property_type": {
32         "varchar": {
33             "max_length": 256
34         }
35     }
36 },
37 {
38     "property_id": 3,
39     "property_name": "type",
40     "property_type": {
41         "varchar": {
42             "max_length": 64
43         }
44     }
45 },
46 ],
47 "primary_keys": [
48     "id"
49 ],
50 },
51 ...
52 ],
53 "edge_types": [
54     {
55         "type_id": 0,
56         "type_name": "HASCREATOR",
57         "vertex_type_pair_relations": [
58             {
59                 "source_vertex": "COMMENT",
60                 "destination_vertex": "PERSON",
61                 "relation": "MANY_TO_ONE",
62                 "x_csr_params": {
63                     "sort_on_compaction": true
64                 }
65             },
66             {
67                 "source_vertex": "POST",
68                 "destination_vertex": "PERSON",
69                 "relation": "MANY_TO_ONE",
70                 "x_csr_params": {
71                     "sort_on_compaction": true
72                 }
73             }
74         ],
75         "properties": [
76             {
77                 "property_id": 0,
78                 "property_name": "creationDate",
79                 "property_type": {
80                     "date": "date"

```

```

81         }
82     }
83 ]
84 },
85 ...
86 ]
87 }
88 }

```

Listing 2.2: Excerpt from the mapping JSON for SF100, describing the data mapping

```

1 {
2   "graph": "ldbc_snb",
3   "loading_config": {
4     "data_source": {
5       "scheme": "file",
6       "location": "./data/social_network-sf100-CsvComposite-StringDateFormatter"
7     },
8     "import_option": "init",
9     "format": {
10      "type": "csv",
11      "metadata": {
12        "delimiter": "|"
13      }
14    }
15  },
16  "vertex_mappings": [
17    {
18      "type_name": "PLACE",
19      "inputs": [
20        "static/place_0_0.csv"
21      ]
22    },
23    {
24      "type_name": "PERSON",
25      "inputs": [
26        "dynamic/person_0_0.csv"
27      ]
28    },
29    ...
30  ],
31  "edge_mappings": [
32    {
33      "type_triplet": {
34        "edge": "HASCREATOR",
35        "source_vertex": "COMMENT",
36        "destination_vertex": "PERSON"
37      },
38      "inputs": [
39        "dynamic/comment_hasCreator_person_0_0.csv.creation_date"
40      ],
41      "source_vertex_mappings": [
42        {
43          "column": {
44            "index": 0,
45            "name": "id"
46          }
47        }
48      ]
49    }
50  ]
51 }

```

```

48     ],
49     "destination_vertex_mappings": [
50         {
51             "column": {
52                 "index": 1,
53                 "name": "id"
54             }
55         }
56     ],
57     "column_mappings": [
58         {
59             "column": {
60                 "index": 2,
61                 "name": "creationDate"
62             },
63             "property": "creationDate"
64         }
65     ]
66 },
67 ...
68 ]
69 }

```

Data loading times are shown for each scale factor in the table below. Values are taken from the database system logs found in `/galaxybase/graph/logs/sys/sys.log`. The column **CSV loading time** shows how long it took to create a graph from the input CSV files. The NA values in the **Data preprocessing time** column are indicating that there were no materialized views or other auxiliary data structures precomputed prior to executing the benchmark, and index creation times are included in the CSV loading times. The column **Total time** contains the sum of the CSV loading and data preprocessing times.

Table 2.2: Data loading times

Scale factor	CSV loading time (s)	Data preprocessing time (s)	Total time (s)
100	437	NA	437
300	1 660	NA	1 660
1000	4 131	NA	4 131

Test Driver Details

3 TEST DRIVER DETAILS

The driver and implementations version used are described below as well as the amount of threads used by the driver for each scale factor.

Table 3.1: Summary of test artifacts and main configuration parameters

Driver version	v1.2.0	https://github.com/ldbc/ldbc_snb_interactive_driver/releases/tag/v1.2.0
Implementations version	v1.0.0	https://github.com/ldbc/ldbc_snb_interactive_impls/releases/tag/1.0.0
LDBC SNB specification version	v0.3.6	https://arxiv.org/pdf/2001.02299v3.pdf
Driver thread count (SF100)	96	
Driver thread count (SF300)	96	
Driver thread count (SF1000)	64	

3.1 Driver implementation

A test driver adaptation for the SUT was provided by the test sponsor. The archive created from the version of the driver used for the audited run is included in the attachments of this report.

The SUT-specific test driver class

`org.ldbcouncil.snb.impls.workloads.galaxybase.interactive.GalaxybaseInteractiveDb` extends the class `org.ldbcouncil.snb.impls.workloads.galaxybase.GalaxybaseDb`, which in turn extends the class `org.ldbcouncil.snb.driver.Db` provided in the LDBC SNB Interactive driver package. Internally, the `GalaxybaseDb` relies on the `OkHttp` client library to issue HTTP requests to communicate with the SUT using local sockets.

3.2 Benchmark configuration of driver

The driver applied the following time compression ratio values:

- TCR=0.001 for scale factor 100,
- TCR=0.00305 for scale factor 300,
- TCR=0.0215 for scale factor 1000,

The complete configuration files for the different scale factors are shown in Listings A.9–A.11, and are also included in the attached supplementary materials.

4 PERFORMANCE RESULTS

The performance results reported here show benchmark executions with scale factors 100, 300, and 1000. The performance summary tables below highlight key performance characteristics. In each case, the query on-time compliance is higher than the minimum required 95% ¹. The performance summary tables below highlight key performance characteristics.

Table 4.1: Summary of results for scale factor 100

Benchmark duration	Benchmark operations	Throughput	Query on-time compliance
02h 03m 56.423s	1 029 999 274	138 507.35 operations second	98.92%

Table 4.2: Summary of results for scale factor 300

Benchmark duration	Benchmark operations	Throughput	Query on-time compliance
02h 03m 15.937s	1 090 027 736	147 381.97 operations second	99.13%

Table 4.3: Summary of results for scale factor 1000

Benchmark duration	Benchmark operations	Throughput	Query on-time compliance
02h 01m 17.610s	984 914 443	135 334.88 operations second	100.00%

¹The total number of late operations for each run in the results in the attachment is referred to as excessive_delay_count.

Performance Results

Table 4.4: Detailed performance benchmark results for scale factor 100 in microseconds

Query	Total count	Min.	Max.	Mean	P ₅₀	P ₉₀	P ₉₅	P ₉₉
Complex 1	5 916 020	256	106 276	2 953.59	1 029	1 804	26 020	40 332
Complex 2	4 157 204	45	71 428	235.44	146	348	636	1 992
Complex 3	1 250 541	9 905	90 268	16 884.07	16 137	20 443	21 431	23 227
Complex 4	4 272 681	32	75 896	693.61	561	1 189	1 496	2 687
Complex 5	1 972 006	2 269	271 520	111 447.78	106 348	161 632	175 848	197 704
Complex 6	354 416	73	70 664	1 195.19	1 275	2 542	2 930	3 970
Complex 7	4 047 804	24	71 328	187.29	98	305	587	1 914
Complex 8	30 763 306	22	73 472	200.71	111	320	612	1 973
Complex 9	291 872	2 995	83 756	10 201.09	10 491	14 500	15 679	17 875
Complex 10	3 845 414	1 955	84 844	7 353.59	7 100	9 508	10 368	12 192
Complex 11	6 991 661	69	71 552	239.14	148	356	649	2 018
Complex 12	3 495 830	476	75 476	2 463.44	2 312	3 469	3 969	5 403
Complex 13	8 095 607	23	72 148	499.46	427	849	1 078	2 339
Complex 14	3 139 113	102	145 224	17 099.16	4 393	57 020	64 484	77 304
Short 1	98 648 095	14	56 384	157.00	67	249	561	1 955
Short 2	98 648 095	17	73 864	183.67	95	281	585	1 976
Short 3	98 648 095	16	92 800	305.54	160	700	1 030	2 545
Short 4	98 655 643	14	73 412	158.38	68	253	562	1 953
Short 5	98 655 643	12	71 628	154.43	66	242	549	1 936
Short 6	98 655 643	14	24 633	155.52	68	244	547	1 921
Short 7	98 655 643	14	74 684	161.54	75	252	554	1 921
Update 1	39 316	70	38 868	377.50	287	588	870	2 267
Update 2	36 881 218	16	73 492	191.08	79	325	705	2 421
Update 3	42 211 499	17	71 592	176.87	76	280	633	2 271
Update 4	706 216	28	24 675	189.19	87	300	657	2 282
Update 5	137 868 892	16	71 764	197.07	80	348	740	2 494
Update 6	9 517 363	21	56 738	201.15	101	318	667	2 269
Update 7	30 247 918	27	73 612	200.88	96	317	673	2 336
Update 8	3 366 520	21	72 036	182.96	80	294	651	2 308

Performance Results

Table 4.5: Detailed performance benchmark results for scale factor 300 in microseconds

Query	Total count	Min.	Max.	Mean	P ₅₀	P ₉₀	P ₉₅	P ₉₉
Complex 1	5 482 542	46	108 564	1 643.12	990	1 564	1 850	38 344
Complex 2	3 852 597	41	24 487	245.76	158	407	684	1 564
Complex 3	1 003 846	3 225	201 640	43 514.48	41 734	56 650	61 462	86 616
Complex 4	3 959 614	31	53 122	795.03	685	1 379	1 611	2 431
Complex 5	1 696 978	734	211 944	98 921.62	97 988	127 648	136 840	151 192
Complex 6	245 769	68	26 307	2 274.93	1 694	5 107	5 685	6 720
Complex 7	4 454 566	26	25 982	191.37	104	357	632	1 506
Complex 8	47 515 365	24	51 502	195.32	108	358	641	1 515
Complex 9	202 193	64	61 892	10 081.01	9 930	15 077	16 108	17 998
Complex 10	3 239 684	39	59 940	8 378.24	8 273	10 659	11 435	13 033
Complex 11	5 939 421	60	25 958	266.75	180	437	710	1 587
Complex 12	3 239 684	53	52 092	2 436.88	2 295	3 639	4 118	5 154
Complex 13	7 502 426	32	50 382	793.57	810	1 250	1 460	2 307
Complex 14	2 909 104	136	197 256	31 670.71	6 115	80 848	91 500	109 920
Short 1	114 534 965	13	52 448	166.62	78	321	615	1 513
Short 2	114 534 965	19	52 084	193.87	106	348	640	1 536
Short 3	114 534 965	18	54 260	329.36	186	751	1 088	2 210
Short 4	114 533 329	14	50 904	169.45	80	327	618	1 516
Short 5	114 533 329	14	49 502	163.35	76	310	606	1 502
Short 6	114 533 329	15	50 736	164.94	78	313	606	1 491
Short 7	114 533 329	13	52 122	172.63	87	323	614	1 493
Update 1	32 260	81	10 409	426.24	330	738	994	1 927
Update 2	27 339 610	15	49 578	194.97	89	386	720	1 951
Update 3	43 901 966	17	49 624	196.69	89	394	730	1 968
Update 4	560 909	28	15 560	200.86	99	380	707	1 922
Update 5	84 558 615	16	51 736	190.65	86	374	706	1 932
Update 6	8 154 173	27	418 272	226.45	117	429	775	1 973
Update 7	29 671 593	29	58 968	221.82	110	432	772	2 027
Update 8	2 826 610	22	49 294	191.82	90	366	694	1 908

Performance Results

Table 4.6: Detailed performance benchmark results for scale factor 1000 in microseconds

Query	Total count	Min.	Max.	Mean	P ₅₀	P ₉₀	P ₉₅	P ₉₉
Complex 1	2 603 777	40	111 896	2 308.90	1 041	1 299	1 393	69 580
Complex 2	1 829 682	33	12 355	110.96	108	123	128	144
Complex 3	410 292	47	108 848	7 662.98	6 219	13 810	16 799	25 915
Complex 4	1 880 506	32	11 655	72.33	61	107	123	184
Complex 5	743 937	34	37 804	7 034.83	8 369	9 976	10 926	11 884
Complex 6	85 048	47	26 011	5 601.62	212	12 452	13 151	14 280
Complex 7	2 707 929	33	13 089	61.03	53	76	86	167
Complex 8	67 698 217	31	14 023	61.11	57	69	75	97
Complex 9	70 008	49	27 767	10 725.46	10 032	16 576	17 736	19 868
Complex 10	1 440 388	47	37 254	9 607.79	9 679	12 807	13 752	15 608
Complex 11	2 603 778	52	154 440	162.01	154	197	226	281
Complex 12	1 538 596	44	17 988	2 492.67	2 455	3 469	3 763	4 399
Complex 13	3 563 064	33	13 504	261.98	208	434	579	1 111
Complex 14	1 381 597	32	598 144	32 709.40	8 101	85 624	105 520	144 296
Short 1	111 142 978	28	13 830	47.32	44	50	54	74
Short 2	111 142 978	30	16 628	64.86	62	70	73	93
Short 3	111 142 978	29	82 984	173.89	118	285	593	957
Short 4	111 145 657	27	14 470	47.99	44	51	54	75
Short 5	111 145 657	27	13 589	46.21	43	49	53	72
Short 6	111 145 657	27	14 325	47.24	44	50	54	73
Short 7	111 145 657	27	17 221	52.22	49	62	66	82
Update 1	12 707	65	7 408	337.64	251	691	960	1 395
Update 2	15 580 659	31	14 120	58.91	48	57	63	148
Update 3	31 167 021	30	13 856	59.00	48	57	62	154
Update 4	220 652	37	11 162	64.54	54	63	70	143
Update 5	34 072 913	30	14 179	58.40	48	56	62	137
Update 6	5 136 214	38	34 644	124.75	64	103	485	1 445
Update 7	30 914 295	36	30 063	68.81	56	68	76	183
Update 8	1 241 601	32	11 902	60.38	50	61	68	137



5 VALIDATION OF THE RESULTS

The scale factor 10 data set was used for validating the correctness of the implementation over the SUT. The validation data set was downloaded from the LDBC repository¹ and it was originally created with the SNB Interactive reference implementation over Neo4j. The system with the driver configuration shown in Listing A.12 successfully returned the expected result sets for the benchmark queries.

¹https://datasets.ldbcouncil.org/interactive-v1/validation_params-interactive-v1.0.0-sf0.1-to-sf10.tar.zst

6 ACID COMPLIANCE

6.1 Transaction isolation level

The benchmark was executed using the *serializable* isolation level setting of the SUT, which is more strict than the *read committed* isolation level minimally required by the SNB Interactive specification, and therefore complies with the requirements of the benchmark.

6.2 SNB ACID test results

The ACID test implementations were reviewed to conform to the ACID test specifications, with all specified test cases implemented. Furthermore, test execution was successful, no atomicity and isolation test failed with serializable isolation level transaction settings.

6.3 Recovery and durability

6.3.1 Recovery

Durability tests were using the regular benchmark workload with scale factor 100, and the server machine was shut down using the command `sleep 2h && sudo reboot`. This command was issued right after the start of the benchmark run. The database server process was manually started again after the crash and it was ready in less than a minute, which was not different from a regular startup time for the server.

6.3.2 Durability

From the driver log, the last update operations before the crash were obtained using the commands below.

```

1 $ grep LdbcUpdate1 LDBC-SNB-results_log.csv | tail -n 1
2 LdbcUpdate1AddPerson|1790757133979|1790757133979|1178|0|1354212945682
3 $ grep LdbcUpdate2 LDBC-SNB-results_log.csv | tail -n 1
4 LdbcUpdate2AddPostLike|1790757134015|1790757134016|143|0|1354212981328
5 $ grep LdbcUpdate3 LDBC-SNB-results_log.csv | tail -n 1
6 LdbcUpdate3AddCommentLike|1790757134015|1790757134016|176|0|1354212981822
7 $ grep LdbcUpdate4 LDBC-SNB-results_log.csv | tail -n 1
8 LdbcUpdate4AddForum|1790757133989|1790757133989|67|0|1354212955682
9 $ grep LdbcUpdate5 LDBC-SNB-results_log.csv | tail -n 1
10 LdbcUpdate5AddForumMembership|1790757134015|1790757134016|84|0|1354212981797
11 $ grep LdbcUpdate6 LDBC-SNB-results_log.csv | tail -n 1
12 LdbcUpdate6AddPost|1790757134010|1790757134010|352|0|1354212975960
13 $ grep LdbcUpdate7 LDBC-SNB-results_log.csv | tail -n 1
14 LdbcUpdate7AddComment|1790757134015|1790757134015|144|0|1354212982190
15 $ grep LdbcUpdate8 LDBC-SNB-results_log.csv | tail -n 1
16 LdbcUpdate8AddFriendship|1790757134009|1790757134009|67|0|1354212975026

```

From the logs, the last completed updates were retrieved for each update operation. The log entries include the operation name, actual and scheduled start time, the execution time, the delay between scheduled and actual start times, and the initial query start time without the TCR multiplier (this latter one is included in the last column). Using this information, the query parameters were obtained from the initial CSV files generated by the Datagen as shown below.

```

1 $ grep -rnw /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/ -e
   '^1354212945682|.|*|1|.*'

```

```

2 /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/updateStream_0_0_person.csv
  :35542:1354212945682|0|1|37383395783526|Peng|Zhang|female|365126400000|1354212945682|27.54.158.160|Internet
  Explorer|322|zh;en|Peng37383395783526@gmail.com;Peng37383395783526@zoho.com
  |1403;1667;2020;2780;3080|2212,2002|937,2004;938,2004
3 $ grep -rnw /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/ -e
  '^1354212981328|.|2|.|.*'
4 /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/updateStream_0_0_forum.csv
  :202408437:1354212981328|1351474607528|2|35184372412757|74766993736581|1354212981328
5 $ grep -rnw /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/ -e
  '^1354212981822|.|3|.|.*'
6 /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/updateStream_0_0_forum.csv
  :202408454:1354212981822|1308021304958|3|17592186293198|74766925009635|1354212981822
7 $ grep -rnw /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/ -e
  '^1354212955682|.|4|.|.*'
8 /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/updateStream_0_0_forum.csv
  :202407446:1354212955682|1354212945682|4|74766833719009|Wall of Peng Zhang
  |1354212955682|37383395783526|1403;1667;2020;2780;3080
9 $ grep -rnw /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/ -e
  '^1354212981797|.|5|.|.*'
10 /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/updateStream_0_0_forum.csv
  :202408451:1354212981797|1336311341353|5|70368763077189|28587302601963|1354212981797
11 $ grep -rnw /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/ -e
  '^1354212975960|.|6|.|.*'
12 /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/updateStream_0_0_forum.csv
  :202408227:1354212975960|1300593596528|6|74767141348948|photo74767141348948.jpg|1354212975960|192.52.232.43|
  Firefox||0|15393163028448|74766805452641|60|
13 $ grep -rnw /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/ -e
  '^1354212982190|.|7|.|.*'
14 /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/updateStream_0_0_forum.csv
  :202408473:1354212982190|1334284314955|7|74766974193918|1354212982190|31.41.41.148|Internet Explorer|fine
  |4|28587302503503|72|74766974193910|-1|
15 $ grep -rnw /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/ -e
  '^1354212975026|.|8|.|.*'
16 /galaxybase/graph/data/social_network-sf100-CsvComposite-StringDateFormatter/updateStream_0_0_forum.csv
  :202408186:1354212975026|1351810013141|8|2199023397680|35184372406626|1354212975026

```

To check whether the graph entities in the driver log entries were persisted in the database, read queries targeting the last completed update operations were executed after database restart. The queries returned the data that was committed according to the logs, so the system passed this check. These queries and their output is shown below using the Cypher REPL of the system (output is slightly formatted for better presentation in this document).

```

1 graph-shell@ldbc_snb> MATCH (v:PERSON) WHERE v.id = "37383395783526" RETURN v;
2 node<1099512111943, PERSON, map[birthday:1981-07-28 browserUsed:Internet Explorer creationDate:1354212945682
  email: Peng37383395783526@gmail.com;Peng37383395783526@zoho.com firstName: Peng gender: female id: 37383395783526
  language: zh;en lastName: Zhang locationIP: 27.54.158.160]>
3 graph-shell@ldbc_snb> MATCH (v:POST)<-[e:LIKES]-(p:PERSON) WHERE v.id = "74766993736581" AND p.id =
  "35184372412757" RETURN e;
4 relationship<0000010000072A2800000090000030003F25A81FFFFFFFFF0000002E, LIKES, from: 1099512097320, to:
  3298601097857, map[creationDate:1354212981328]>
5 graph-shell@ldbc_snb> MATCH (v:COMMENT)<-[e:LIKES]-(p:PERSON) WHERE v.id = "74766925009635" AND p.id =
  "17592186293198" RETURN e;
6 relationship<0000010000060F950000009000002000EB4E795FFFFFFFFF00000056, LIKES, from: 1099512024981, to:
  2199269992341, map[creationDate:1354212981822]>
7 graph-shell@ldbc_snb> MATCH (v:FORUM) WHERE v.id = "74766833719009" RETURN v;
8 node<4398051216234, FORUM, map[creationDate:1354212955682 id:74766833719009 title:Wall of Peng Zhang]>
9 graph-shell@ldbc_snb> MATCH (p:PERSON)<-[e:HASMEMBER]-(f:FORUM) WHERE p.id = "28587302601963" AND f.id =
  "70368763077189" RETURN e;

```



```

10 relationship<00000400004075410000000400000100000031A8FFFFFFFF00000862, HASMEMBER, from: 4398050735425, to:
    1099511640488, map[joinDate:1354212981797]>
11 graph-shell@ldbc_snb> MATCH (v:POST) WHERE v.id = "74767141348948" RETURN v;
12 node<3298601383654, POST, map[browserUsed:Firefox content: creationDate:1354212975960 id:74767141348948 imageFile
    :photo74767141348948.jpg language: length:0 locationIP:192.52.232.43]>
13 graph-shell@ldbc_snb> MATCH (v:COMMENT) WHERE v.id = "74766974193918" RETURN v;
14 node<2199270818316, COMMENT, map[browserUsed:Internet Explorer content:fine creationDate:1354212982190 id
    :74766974193918 length:4 locationIP:31.41.41.148]>
15 graph-shell@ldbc_snb> MATCH (p2:PERSON)<-[e:KNOWS]-(p1:PERSON) WHERE p2.id = "35184372406626" AND p1.id =
    "2199023397680" RETURN e;
16 relationship<000001000003EAA5000000080000010000073117FFFFFFFF000000BC, KNOWS, from: 1099511884453, to:
    1099512099095, map[creationDate:1354212975026]>

```

6.3.3 Consistency after recovery

The benchmark relied on no precomputed auxiliary data structures, thus no additional steps were required to check consistency. After the crash, the database was successfully started using the usual startup procedure.

Supplementary Materials

7 SUPPLEMENTARY MATERIALS

The table below shows the list of supplementary materials. These materials are made available with this full disclosure report to allow reproducibility of results.

Table 7.1: Supplementary materials

File	Purpose
README.md	Benchmarking README
results-sf{100,300,1000}.tar.gz	Driver output files for the selected scale factors
galaxybase-amd64-3.5.3- [...].tar.gz	Galaxybase database distribution used for the benchmark
OpenJDK17U- [...].tar.gz	OpenJDK 17 distribution used to run the benchmark components
galaxybase-ldbc-driver-main	Galaxybase-specific LDBC SNB driver source code
benchmark-sf{100,300,1000}.properties	Driver configurations for the selected scale factors
validate.properties	Driver configuration for validating benchmark results
run_driver.sh	Script to run the LDBC SNB benchmark driver
procedures	Stored-procedure implementations of the LDBC SNB queries
«snb_interactive»procedure.java	Interactive workload query procedure implementations
AcidProcedure.java	ACID test procedure implementation
schema	Graph schema and data mapping configuration
schema/schema.json	Galaxybase graph schema definition
schema/mapping.json	Mapping of LDBC SNB data to the graph schema
load_graph.sh	Script to create and load the benchmark graph
add-numactl.sh	Script to configure NUMA-related execution settings
remove-numactl.sh	Script to restore the original NUMA-related execution settings
AddColumn.java	Utility for preparing the benchmark data for loading

The attachment folder directory structure is as follows:

```

attachments
├── galaxybase_ldbc
│   ├── add-numactl.sh
│   ├── AddColumn.java
│   ├── driver
│   │   ├── benchmark-sf{100,300,1000}.properties
│   │   ├── data
│   │   └── galaxybase-ldbc-driver-main
│   ├── run_driver.sh
│   └── validate.properties
├── galaxybase-amd64-3.5.3-enterprise-undocker-20260923141832.tar.gz
├── load_graph.sh
├── OpenJDK17U-jdk_x64_linux_hotspot_17.0.6_10.tar.gz
├── procedures
│   ├── AcidProcedure.java
│   └── «snb_interactive»Procedure.java
├── README.md
├── remove-numactl.sh
├── schema
│   ├── mapping.json
│   └── schema.json
├── results
│   └── results-sf{100,300,1000}.zip

```

A APPENDIX

A.1 cpu details

Listing A.1: output of the cat /proc/cpuinfo command for one core

```
1 processor : 0
2 vendor_id : authenticamd
3 cpu family : 26
4 model : 2
5 model name : amd epyc 9r05 64-core processor
6 stepping : 1
7 microcode : 0xb002161
8 cpu mhz : 2396.429
9 cache size : 1024 kb
10 physical id : 0
11 siblings : 48
12 core id : 0
13 cpu cores : 48
14 apicid : 0
15 initial apicid : 0
16 fpu : yes
17 fpu_exception : yes
18 cpuid level : 16
19 wp : yes
20 flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht
    syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good amd_lbr_v2 nopl xtopology nonstop_tsc
    cpuid extd_apicid aperfmperf rapl pni pclmulqdq monitor ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt
    aes xsave avx f16c rdrand lahf_lm cmp_legacy svm extapic cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw
    ibs skinit wdt tce topoext perfctr_core perfctr_nb bpext perfctr_llc mwaitx cpuid_fault cpb cat_l3 cdp_l3
    hw_pstate ssbd mba perfmon_v2 ibrs ibpb stibp ibrs_enhanced vmmcall fsgsbase tsc_adjust bmi1 avx2 smep bmi2
    erms invpcid cqm rdt_a avx512f avx512dq rdseed adx smap avx512ifma clflushopt clwb avx512cd sha_ni avx512bw
    avx512vl xsaveopt xsavec xgetbv1 xsaves cqm_llc cqm_occup_llc cqm_mbm_total cqm_mbm_local user_shstk avx_vnni
    avx512_bf16 clzero irperf xsaveerptr rdpru wbnoinvd amd_ppin cppc arat npt lbrv svm_lock nrip_save tsc_scale
    vmcb_clean flushbyasid decodeassists pausefilter pfthreshold avic v_vmsave_vmload vgif x2avic v_spec_ctrl
    vnmi avx512vbmi umip pku ospke avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq
    la57 rdpid bus_lock_detect movdiri movdir64b overflow_recov succor smca fsrm avx512_vp2intersect flush_l1d
    sev sev_es debug_swap amd_lbr_pmc_freeze
21 bugs : sysret_ss_attrs spectre_v1 spectre_v2 spec_store_bypass srso spectre_v2_user vmscape
22 bogomips : 6600.48
23 tlb size : 192 4k pages
24 clflush size : 64
25 cache_alignment : 64
26 address sizes : 46 bits physical, 57 bits virtual
27 power management: ts ttp tm hwpstate cpb eff_freq_ro [13] [14]
```

Listing A.2: output of the lscpu command

```
1 architecture: x86_64
2 cpu op-mode(s): 32-bit, 64-bit
3 address sizes: 46 bits physical, 57 bits virtual
4 byte order: little endian
5 cpu(s): 96
6 on-line cpu(s) list: 0-95
7 vendor id: authenticamd
8 model name: amd epyc 9r05 64-core processor
9 cpu family: 26
```



appendix

A.1. cpu details

```

10  model:                2
11  thread(s) per core:   1
12  core(s) per socket:   48
13  socket(s):            2
14  stepping:            1
15  frequency boost:      enabled
16  cpu(s) scaling mhz:    53%
17  cpu max mhz:          5008.9292
18  cpu min mhz:          1217.8580
19  bogomips:             6600.48
20  flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx
    fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc
21                        rep_good amd_lbr_v2 nopl xtopology nonstop_tsc cpuid extd_apicid aperfmperf rapl pni
    pclmulqdq monitor ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt ae
22                        s xsave avx f16c rdrand lahf_lm cmp_legacy svm extapic cr8_legacy abm sse4a
    misalignsse 3dnowprefetch osvw ibs skinit wdt tce topoext perfctr_core perfctr_nb b
23                        pext perfctr_llc mwaitx cpuid_fault cpb cat_l3 cdp_l3 hw_pstate ssbd mba perfmon_v2
    ibrs ibpb stibp ibrs_enhanced vmmcall fsgsbase tsc_adjust bmi1 avx2 smep bm
24                        i2 erms invpcid cqm rdt_a avx512f avx512dq rdseed adx smap avx512ifma clflushopt
    clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves cqm_llc
25                        cqm_occup_llc cqm_mbm_total cqm_mbm_local user_shstk avx_vnni avx512_bf16 clzero
    irperf xsaveerptr rdpru wbnoinvd amd_ppin cppc arat npt lbrv svm_lock nrip_sav
26                        e tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold avic
    v_umsave_vmload vgif x2avic v_spec_ctrl vnmi avx512vbmi umip pku ospke avx512_vbm
27                        i2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq la57 rdpid
    bus_lock_detect movdiri movdir64b overflow_recov succor smca fsrm avx512_vp2inter
28                        sect flush_l1d sev sev_es debug_swap amd_lbr_pmc_freeze
29  virtualization features:
30      virtualization:    amd-v
31  caches (sum of all):
32      l1d:               4.5 mib (96 instances)
33      l1i:               3 mib (96 instances)
34      l2:                96 mib (96 instances)
35      l3:                512 mib (16 instances)
36  numa:
37      numa node(s):       2
38      numa node0 cpu(s):  0-47
39      numa node1 cpu(s):  48-95
40  vulnerabilities:
41      gather data sampling: not affected
42      ghostwrite:          not affected
43      indirect target selection: not affected
44      itlb multihit:       not affected
45      l1tf:                not affected
46      mds:                 not affected
47      meltdown:            not affected
48      mmio stale data:     not affected
49      old microcode:       not affected
50      reg file data sampling: not affected
51      retbleed:            not affected
52      spec rstack overflow: mitigation; reduced speculation
53      spec store bypass:   mitigation; speculative store bypass disabled via prctl
54      spectre v1:          mitigation; usercopy/swaps barriers and __user pointer sanitization
55      spectre v2:          mitigation; enhanced / automatic ibrs; ibpb conditional; stibp disabled; pbrsb-eibrs
    not affected; bhi not affected
56      srbds:               not affected
57      tsa:                 not affected
58      tsx async abort:     not affected

```



```
59 vmscape: mitigation; ibpb before exit to userspace
```

A.2 memory details

Listing A.3: output of the `lshw -c memory` command

```

1  *-cache:0
2      description: l1 cache
3      physical id: 1b
4      slot: l1 - cache
5      size: 3840kib
6      capacity: 3840kib
7      clock: 1ghz (1.0ns)
8      capabilities: pipeline-burst internal write-back unified
9      configuration: level=1
10 *-cache:1
11     description: l2 cache
12     physical id: 1c
13     slot: l2 - cache
14     size: 48mib
15     capacity: 48mib
16     clock: 1ghz (1.0ns)
17     capabilities: pipeline-burst internal write-back unified
18     configuration: level=2
19 *-cache:2
20     description: l3 cache
21     physical id: 1d
22     slot: l3 - cache
23     size: 256mib
24     capacity: 256mib
25     clock: 1ghz (1.0ns)
26     capabilities: pipeline-burst internal write-back unified
27     configuration: level=3
28 *-cache:3
29     description: l1 cache
30     physical id: 1e
31     slot: l1 - cache
32     size: 3840kib
33     capacity: 3840kib
34     clock: 1ghz (1.0ns)
35     capabilities: pipeline-burst internal write-back unified
36     configuration: level=1
37 *-cache:4
38     description: l2 cache
39     physical id: 1f
40     slot: l2 - cache
41     size: 48mib
42     capacity: 48mib
43     clock: 1ghz (1.0ns)
44     capabilities: pipeline-burst internal write-back unified
45     configuration: level=2
46 *-cache:5
47     description: l3 cache
48     physical id: 20
49     slot: l3 - cache
50     size: 256mib

```



```

51     capacity: 256mib
52     clock: 1ghz (1.0ns)
53     capabilities: pipeline-burst internal write-back unified
54     configuration: level=3
55 *--memory
56     description: system memory
57     physical id: 26
58     slot: system board or motherboard
59     size: 3tib
60     capacity: 12tib
61     capabilities: ecc
62     configuration: errordetection=multi-bit-ecc
63 *--bank:0
64     description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
65     product: m321raja0mb2-ccpkc
66     vendor: samsung
67     physical id: 0
68     serial: 2606-02b94872
69     slot: cpu0 channel0 dimm0
70     size: 128gib
71     width: 64 bits
72     clock: 2105mhz (0.5ns)
73 *--bank:1
74     description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
75     product: m321raja0mb2-ccpkc
76     vendor: samsung
77     physical id: 1
78     serial: 2606-02b9485d
79     slot: cpu0 channel1 dimm0
80     size: 128gib
81     width: 64 bits
82     clock: 2105mhz (0.5ns)
83 *--bank:2
84     description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
85     product: m321raja0mb2-ccpkc
86     vendor: samsung
87     physical id: 2
88     serial: 2606-02b94734
89     slot: cpu0 channel2 dimm0
90     size: 128gib
91     width: 64 bits
92     clock: 2105mhz (0.5ns)
93 *--bank:3
94     description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
95     product: m321raja0mb2-ccpkc
96     vendor: samsung
97     physical id: 3
98     serial: 2606-02b9486a
99     slot: cpu0 channel3 dimm0
100    size: 128gib
101    width: 64 bits
102    clock: 2105mhz (0.5ns)
103 *--bank:4
104    description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
105    product: m321raja0mb2-ccpkc
106    vendor: samsung
107    physical id: 4
108    serial: 2606-02b947a9

```

appendix

A.2. memory details

```

109         slot: cpu0 channel4 dimm0
110         size: 128gib
111         width: 64 bits
112         clock: 2105mhz (0.5ns)
113     *--bank:5
114         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
115         product: m321raja0mb2-ccpkc
116         vendor: samsung
117         physical id: 5
118         serial: 2606-02b9482f
119         slot: cpu0 channel5 dimm0
120         size: 128gib
121         width: 64 bits
122         clock: 2105mhz (0.5ns)
123     *--bank:6
124         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
125         product: m321raja0mb2-ccpkc
126         vendor: samsung
127         physical id: 6
128         serial: 2606-02b94399
129         slot: cpu0 channel6 dimm0
130         size: 128gib
131         width: 64 bits
132         clock: 2105mhz (0.5ns)
133     *--bank:7
134         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
135         product: m321raja0mb2-ccpkc
136         vendor: samsung
137         physical id: 7
138         serial: 2606-02b94416
139         slot: cpu0 channel7 dimm0
140         size: 128gib
141         width: 64 bits
142         clock: 2105mhz (0.5ns)
143     *--bank:8
144         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
145         product: m321raja0mb2-ccpkc
146         vendor: samsung
147         physical id: 8
148         serial: 2606-02b94653
149         slot: cpu0 channel8 dimm0
150         size: 128gib
151         width: 64 bits
152         clock: 2105mhz (0.5ns)
153     *--bank:9
154         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
155         product: m321raja0mb2-ccpkc
156         vendor: samsung
157         physical id: 9
158         serial: 2606-02b94373
159         slot: cpu0 channel9 dimm0
160         size: 128gib
161         width: 64 bits
162         clock: 2105mhz (0.5ns)
163     *--bank:10
164         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
165         product: m321raja0mb2-ccpkc
166         vendor: samsung

```



appendix

A.2. memory details

```

167     physical id: a
168     serial: 2606-02b943db
169     slot: cpu0 channel10 dimm0
170     size: 128gib
171     width: 64 bits
172     clock: 2105mhz (0.5ns)
173     *-bank:11
174         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
175         product: m321raja0mb2-ccpkc
176         vendor: samsung
177         physical id: b
178         serial: 2606-02b9429b
179         slot: cpu0 channel11 dimm0
180         size: 128gib
181         width: 64 bits
182         clock: 2105mhz (0.5ns)
183     *-bank:12
184         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
185         product: m321raja0mb2-ccpkc
186         vendor: samsung
187         physical id: c
188         serial: 2606-02b9482e
189         slot: cpu1 channel0 dimm0
190         size: 128gib
191         width: 64 bits
192         clock: 2105mhz (0.5ns)
193     *-bank:13
194         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
195         product: m321raja0mb2-ccpkc
196         vendor: samsung
197         physical id: d
198         serial: 2606-02b94692
199         slot: cpu1 channel1 dimm0
200         size: 128gib
201         width: 64 bits
202         clock: 2105mhz (0.5ns)
203     *-bank:14
204         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
205         product: m321raja0mb2-ccpkc
206         vendor: samsung
207         physical id: e
208         serial: 2606-02b98251
209         slot: cpu1 channel2 dimm0
210         size: 128gib
211         width: 64 bits
212         clock: 2105mhz (0.5ns)
213     *-bank:15
214         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
215         product: m321raja0mb2-ccpkc
216         vendor: samsung
217         physical id: f
218         serial: 2606-02b942dc
219         slot: cpu1 channel3 dimm0
220         size: 128gib
221         width: 64 bits
222         clock: 2105mhz (0.5ns)
223     *-bank:16
224         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)

```



appendix

A.2. memory details

```

225     product: m321raja0mb2-ccpkc
226     vendor: samsung
227     physical id: 10
228     serial: 2606-02b9491a
229     slot: cpu1 channel4 dimm0
230     size: 128gib
231     width: 64 bits
232     clock: 2105mhz (0.5ns)
233     *-bank:17
234         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
235         product: m321raja0mb2-ccpkc
236         vendor: samsung
237         physical id: 11
238         serial: 2606-02b980d0
239         slot: cpu1 channel5 dimm0
240         size: 128gib
241         width: 64 bits
242         clock: 2105mhz (0.5ns)
243     *-bank:18
244         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
245         product: m321raja0mb2-ccpkc
246         vendor: samsung
247         physical id: 12
248         serial: 2606-02b981f5
249         slot: cpu1 channel6 dimm0
250         size: 128gib
251         width: 64 bits
252         clock: 2105mhz (0.5ns)
253     *-bank:19
254         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
255         product: m321raja0mb2-ccpkc
256         vendor: samsung
257         physical id: 13
258         serial: 2606-02b97907
259         slot: cpu1 channel7 dimm0
260         size: 128gib
261         width: 64 bits
262         clock: 2105mhz (0.5ns)
263     *-bank:20
264         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
265         product: m321raja0mb2-ccpkc
266         vendor: samsung
267         physical id: 14
268         serial: 2606-02b947c9
269         slot: cpu1 channel8 dimm0
270         size: 128gib
271         width: 64 bits
272         clock: 2105mhz (0.5ns)
273     *-bank:21
274         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
275         product: m321raja0mb2-ccpkc
276         vendor: samsung
277         physical id: 15
278         serial: 2606-02b981be
279         slot: cpu1 channel9 dimm0
280         size: 128gib
281         width: 64 bits
282         clock: 2105mhz (0.5ns)

```



```

283     *-bank:22
284         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
285         product: m321raja0mb2-ccpkc
286         vendor: samsung
287         physical id: 16
288         serial: 2606-02b94870
289         slot: cpu1 channel10 dimm0
290         size: 128gib
291         width: 64 bits
292         clock: 2105mhz (0.5ns)
293     *-bank:23
294         description: dimm synchronous registered (buffered) 6400 mhz (0.2 ns)
295         product: m321raja0mb2-ccpkc
296         vendor: samsung
297         physical id: 17
298         serial: 2606-02b94858
299         slot: cpu1 channel11 dimm0
300         size: 128gib
301         width: 64 bits
302         clock: 2105mhz (0.5ns)
303     *-firmware
304         description: bios
305         vendor: amazon ec2
306         physical id: 73
307         version: 1.0
308         date: 10/16/2017
309         size: 64kib
310         capacity: 64mib
311         capabilities: pci pnp upgrade shadowing bootselect edd int5printscreen int9keyboard int14serial
int17printer int10video acpi usb biosbootspecification netboot uefi

```

A.3 storage details and io performance

Listing A.4: output of the `lshw -c storage -c disk` command

```

1     *-nvme
2         description: nvme device
3         product: amazon elastic block store
4         vendor: amazon.com, inc.
5         physical id: 0
6         bus info: pci@0000:05:00.0
7         logical name: /dev/nvme0
8         version: 2.0
9         serial: vol05ae04f8b5a376e45
10        width: 64 bits
11        clock: 33mhz
12        capabilities: nvme pm pciexpress msix nvm_express bus_master cap_list
13        configuration: driver=nvme latency=0 nqn=nqn:2008-08.com.amazon.aws:ebs:vol05ae04f8b5a376e45 state=live
14        resources: irq:0 memory:a0200000-a02fffff
15     *-sata:0
16         description: sata controller
17         product: fch sata controller [ahci mode]
18         vendor: advanced micro devices, inc. [amd]
19         physical id: 0
20         bus info: pci@0000:66:00.0
21         version: 93

```



```

22     width: 32 bits
23     clock: 33mhz
24     capabilities: sata pm pciexpress msi msix ahci_1.0 bus_master cap_list
25     configuration: driver=ahci latency=0
26     resources: irq:332 memory:99201000-992017ff
27 *-sata:1
28     description: sata controller
29     product: fch sata controller [ahci mode]
30     vendor: advanced micro devices, inc. [amd]
31     physical id: 0.1
32     bus info: pci@0000:66:00.1
33     version: 93
34     width: 32 bits
35     clock: 33mhz
36     capabilities: sata pm pciexpress msi msix ahci_1.0 bus_master cap_list
37     configuration: driver=ahci latency=0
38     resources: irq:340 memory:99200000-992007ff
39 *-nvme
40     description: nvme device
41     product: amazon ec2 nvme instance storage
42     vendor: amazon.com, inc.
43     physical id: 0
44     bus info: pci@0000:73:00.0
45     logical name: /dev/nvme2
46     version: 0
47     serial: aws0748ac167021ec4e7
48     width: 64 bits
49     clock: 33mhz
50     capabilities: nvme pm pciexpress msix nvmm_express bus_master cap_list
51     configuration: driver=nvme latency=0 nqn=nqn.2008-08.com.amazon.aws:ec2-nvme-instance:aws0748ac167021ec4e7
52     state=live
53     resources: irq:0 memory:f9e00000-f9e03fff
54 *-sata:0
55     description: sata controller
56     product: fch sata controller [ahci mode]
57     vendor: advanced micro devices, inc. [amd]
58     physical id: 0
59     bus info: pci@0000:7e:00.0
60     version: 93
61     width: 32 bits
62     clock: 33mhz
63     capabilities: sata pm pciexpress msi msix ahci_1.0 bus_master cap_list
64     configuration: driver=ahci latency=0
65     resources: irq:342 memory:fcf01000-fcf017ff
66 *-sata:1
67     description: sata controller
68     product: fch sata controller [ahci mode]
69     vendor: advanced micro devices, inc. [amd]
70     physical id: 0.1
71     bus info: pci@0000:7e:00.1
72     version: 93
73     width: 32 bits
74     clock: 33mhz
75     capabilities: sata pm pciexpress msi msix ahci_1.0 bus_master cap_list
76     configuration: driver=ahci latency=0
77     resources: irq:344 memory:fcf00000-fcf007ff
78 *-sata:0
79     description: sata controller

```

```

79     product: fch sata controller [ahci mode]
80     vendor: advanced micro devices, inc. [amd]
81     physical id: 0
82     bus info: pci@0000:e6:00.0
83     version: 93
84     width: 32 bits
85     clock: 33mhz
86     capabilities: sata pm pciexpress msi msix ahci_1.0 bus_master cap_list
87     configuration: driver=ahci latency=0
88     resources: irq:346 memory:f0101000-f01017ff
89 *-sata:1
90     description: sata controller
91     product: fch sata controller [ahci mode]
92     vendor: advanced micro devices, inc. [amd]
93     physical id: 0.1
94     bus info: pci@0000:e6:00.1
95     version: 93
96     width: 32 bits
97     clock: 33mhz
98     capabilities: sata pm pciexpress msi msix ahci_1.0 bus_master cap_list
99     configuration: driver=ahci latency=0
100    resources: irq:348 memory:f0100000-f01007ff
101 *-nvme
102     description: nvme device
103     product: amazon ec2 nvme instance storage
104     vendor: amazon.com, inc.
105     physical id: 0
106     bus info: pci@0000:f3:00.0
107     logical name: /dev/nvme1
108     version: 0
109     serial: aws21e9afeba2c18c8dc
110     width: 64 bits
111     clock: 33mhz
112     capabilities: nvme pm pciexpress msix nvm_express bus_master cap_list
113     configuration: driver=nvme latency=0 nqn=nqn.2008-08.com.amazon.aws:ec2-nvme-instance:aws21e9afeba2c18c8dc
114     state=live
115     resources: irq:0 memory:f4c00000-f4c03fff
116 *-sata:0
117     description: sata controller
118     product: fch sata controller [ahci mode]
119     vendor: advanced micro devices, inc. [amd]
120     physical id: 0
121     bus info: pci@0000:fe:00.0
122     version: 93
123     width: 32 bits
124     clock: 33mhz
125     capabilities: sata pm pciexpress msi msix ahci_1.0 bus_master cap_list
126     configuration: driver=ahci latency=0
127     resources: irq:350 memory:97201000-972017ff
128 *-sata:1
129     description: sata controller
130     product: fch sata controller [ahci mode]
131     vendor: advanced micro devices, inc. [amd]
132     physical id: 0.1
133     bus info: pci@0000:fe:00.1
134     version: 93
135     width: 32 bits
136     clock: 33mhz

```

```

136 capabilities: sata pm pciexpress msi msix ahci_1.0 bus_master cap_list
137 configuration: driver=ahci latency=0
138 resources: irq:352 memory:97200000-972007ff

```

Listing A.5: output of the `lsblk -o name,size,type,fstype,mountpoints,model,serial,tran` command

```

1 name          size type fstype  mountpoints  model  serial  tran
2 loop0         28.2m loop squashfs /snap/amazon-ssm-agent/13009
3 loop1         28.4m loop squashfs /snap/amazon-ssm-agent/13349
4 loop2         74m loop squashfs /snap/core22/2411
5 loop3         74m loop squashfs /snap/core22/2955
6 loop4         49.3m loop squashfs /snap/snapd/26865
7 loop5         50.3m loop squashfs /snap/snapd/27738
8 nvme0n1       100g disk amazon elastic block store      vol05ae04f8b5a376e45 nvme
9 nvme0n1p1    98.9g part ext4      / nvme
10 nvme0n1p13  1023m part ext4      /boot nvme
11 nvme0n1p14   4m part              nvme
12 nvme0n1p15  106m part vfat       /boot/efi nvme
13 nvme2n1       3.5t disk xfs        /galaxybase amazon ec2 nvme instance storage aws0748ac167021ec4e7 nvme
14 nvme1n1       3.5t disk xfs        /driver  amazon ec2 nvme instance storage aws21e9afeba2c18c8dc nvme

```

Listing A.6: output of the `fio` command

```

1 $ fio --rw=write --ioengine=sync --fdatasync=1 --direct=1 --directory=io-test-data --size=2g --bs=4k --name=
  iotest
2 iotest: (g=0): rw=write, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=sync, iodepth=1
3 fio-3.41
4 Starting 1 process
5 Jobs: 1 (f=1): [W(1)][100.0%][w=237MiB/s][w=60.8k IOPS][eta 00m:00s]
6 iotest: (groupid=0, jobs=1): err= 0: pid=44901: Wed Sep 30 20:52:36 2026
7   write: IOPS=60.8k, BW=237MiB/s (249MB/s)(2048MiB/8624msec); 0 zone resets
8     clat (nsec): min=14980, max=172921, avg=16109.78, stdev=573.01
9     lat (nsec): min=15000, max=172941, avg=16127.63, stdev=573.56
10    clat percentiles (nsec):
11      | 1.00th=[15552],  5.00th=[15680], 10.00th=[15808], 20.00th=[15936],
12      | 30.00th=[15936], 40.00th=[16064], 50.00th=[16064], 60.00th=[16064],
13      | 70.00th=[16192], 80.00th=[16192], 90.00th=[16320], 95.00th=[16512],
14      | 99.00th=[17024], 99.50th=[18304], 99.90th=[22400], 99.95th=[23168],
15      | 99.99th=[29056]
16    bw ( KiB/s): min=241816, max=243904, per=100.00%, avg=243210.35, stdev=514.77, samples=17
17    iops        : min=60454, max=60976, avg=60802.59, stdev=128.69, samples=17
18    lat (usec)   : 20=99.61%, 50=0.39%, 100=0.01%, 250=0.01%
19    fsync/fdatasync/sync_file_range:
20      sync (nsec): min=130, max=10020, avg=146.89, stdev=39.17
21      sync percentiles (nsec):
22        | 1.00th=[ 141],  5.00th=[ 141], 10.00th=[ 141], 20.00th=[ 141],
23        | 30.00th=[ 141], 40.00th=[ 151], 50.00th=[ 151], 60.00th=[ 151],
24        | 70.00th=[ 151], 80.00th=[ 151], 90.00th=[ 151], 95.00th=[ 151],
25        | 99.00th=[ 161], 99.50th=[ 161], 99.90th=[ 171], 99.95th=[ 262],
26        | 99.99th=[ 1864]
27    cpu         : usr=3.04%, sys=9.30%, ctx=524290, majf=0, minf=9
28    IO depths    : 1=200.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
29      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
30      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
31      issued rwts: total=0,524288,0,0 short=524287,0,0,0 dropped=0,0,0,0
32      latency    : target=0, window=0, percentile=100.00%, depth=1
33
34 Run status group 0 (all jobs):

```

```

35 WRITE: bw=237MiB/s (249MB/s), 237MiB/s-237MiB/s (249MB/s-249MB/s), io=2048MiB (2147MB), run=8624-8624msec
36
37 Disk stats (read/write):
38 nvme2n1: ios=0/510387, sectors=0/4083244, merge=0/0, ticks=0/7509, in_queue=7509, util=87.47%

```

A.4 import configuration

Listing A.7: content of schema.json describing the data schema

```

1 {
2   "name": "ldbc_snb",
3   "storagev2": true,
4   "storageengine": "storage-v2",
5   "store_type": "mutable_csr",
6   "schema": {
7     "vertex_types": [
8       {
9         "type_id": 0,
10        "type_name": "place",
11        "properties": [
12          {
13            "property_id": 0,
14            "property_name": "id",
15            "property_type": {
16              "primitive_type": "dt_signed_int64"
17            }
18          },
19          {
20            "property_id": 1,
21            "property_name": "name",
22            "property_type": {
23              "varchar": {
24                "max_length": 256
25              }
26            }
27          },
28          {
29            "property_id": 2,
30            "property_name": "url",
31            "property_type": {
32              "varchar": {
33                "max_length": 256
34              }
35            }
36          },
37          {
38            "property_id": 3,
39            "property_name": "type",
40            "property_type": {
41              "varchar": {
42                "max_length": 64
43              }
44            }
45          }
46        ],
47        "primary_keys": [

```

```

48         "id"
49     ]
50 },
51 {
52     "type_id": 1,
53     "type_name": "person",
54     "properties": [
55         {
56             "property_id": 0,
57             "property_name": "id",
58             "property_type": {
59                 "primitive_type": "dt_signed_int64"
60             }
61         },
62         {
63             "property_id": 1,
64             "property_name": "firstname",
65             "property_type": {
66                 "varchar": {
67                     "max_length": 40
68                 }
69             }
70         },
71         {
72             "property_id": 2,
73             "property_name": "lastname",
74             "property_type": {
75                 "varchar": {
76                     "max_length": 40
77                 }
78             }
79         },
80         {
81             "property_id": 3,
82             "property_name": "gender",
83             "property_type": {
84                 "varchar": {
85                     "max_length": 40
86                 }
87             }
88         },
89         {
90             "property_id": 4,
91             "property_name": "birthday",
92             "property_type": {
93                 "day": "day"
94             }
95         },
96         {
97             "property_id": 5,
98             "property_name": "creationdate",
99             "property_type": {
100                 "date": "date"
101             }
102         },
103         {
104             "property_id": 6,
105             "property_name": "locationip",

```

```

106         "property_type": {
107             "varchar": {
108                 "max_length": 40
109             }
110         },
111     },
112     {
113         "property_id": 7,
114         "property_name": "browserused",
115         "property_type": {
116             "varchar": {
117                 "max_length": 40
118             }
119         },
120     },
121     {
122         "property_id": 8,
123         "property_name": "language",
124         "property_type": {
125             "varchar": {
126                 "max_length": 2048
127             }
128         },
129     },
130     {
131         "property_id": 9,
132         "property_name": "email",
133         "property_type": {
134             "varchar": {
135                 "max_length": 2048
136             }
137         },
138     },
139 ],
140 "primary_keys": [
141     "id"
142 ],
143 },
144 {
145     "type_id": 2,
146     "type_name": "comment",
147     "properties": [
148         {
149             "property_id": 0,
150             "property_name": "id",
151             "property_type": {
152                 "primitive_type": "dt_signed_int64"
153             }
154         },
155         {
156             "property_id": 1,
157             "property_name": "creationdate",
158             "property_type": {
159                 "date": "date"
160             }
161         },
162         {
163             "property_id": 2,

```

```

164         "property_name": "locationip",
165         "property_type": {
166             "varchar": {
167                 "max_length": 40
168             }
169         }
170     },
171     {
172         "property_id": 3,
173         "property_name": "browserused",
174         "property_type": {
175             "varchar": {
176                 "max_length": 40
177             }
178         }
179     },
180     {
181         "property_id": 4,
182         "property_name": "content",
183         "property_type": {
184             "varchar": {
185                 "max_length": 2000
186             }
187         }
188     },
189     {
190         "property_id": 5,
191         "property_name": "length",
192         "property_type": {
193             "primitive_type": "dt_signed_int32"
194         }
195     }
196 ],
197 "primary_keys": [
198     "id"
199 ]
200 },
201 {
202     "type_id": 3,
203     "type_name": "post",
204     "properties": [
205         {
206             "property_id": 0,
207             "property_name": "id",
208             "property_type": {
209                 "primitive_type": "dt_signed_int64"
210             }
211         },
212         {
213             "property_id": 1,
214             "property_name": "imagefile",
215             "property_type": {
216                 "varchar": {
217                     "max_length": 40
218                 }
219             }
220         },
221         {

```

```

222         "property_id": 2,
223         "property_name": "creationdate",
224         "property_type": {
225             "date": "date"
226         }
227     },
228     {
229         "property_id": 3,
230         "property_name": "locationip",
231         "property_type": {
232             "varchar": {
233                 "max_length": 40
234             }
235         }
236     },
237     {
238         "property_id": 4,
239         "property_name": "browserused",
240         "property_type": {
241             "varchar": {
242                 "max_length": 40
243             }
244         }
245     },
246     {
247         "property_id": 5,
248         "property_name": "language",
249         "property_type": {
250             "varchar": {
251                 "max_length": 40
252             }
253         }
254     },
255     {
256         "property_id": 6,
257         "property_name": "content",
258         "property_type": {
259             "varchar": {
260                 "max_length": 2000
261             }
262         }
263     },
264     {
265         "property_id": 7,
266         "property_name": "length",
267         "property_type": {
268             "primitive_type": "dt_signed_int32"
269         }
270     }
271 ],
272 "primary_keys": [
273     "id"
274 ]
275 },
276 {
277     "type_id": 4,
278     "type_name": "forum",
279     "properties": [

```

```

280         {
281             "property_id": 0,
282             "property_name": "id",
283             "property_type": {
284                 "primitive_type": "dt_signed_int64"
285             }
286         },
287         {
288             "property_id": 1,
289             "property_name": "title",
290             "property_type": {
291                 "varchar": {
292                     "max_length": 256
293                 }
294             }
295         },
296         {
297             "property_id": 2,
298             "property_name": "creationdate",
299             "property_type": {
300                 "date": "date"
301             }
302         }
303     ],
304     "primary_keys": [
305         "id"
306     ]
307 },
308 {
309     "type_id": 5,
310     "type_name": "organisation",
311     "properties": [
312         {
313             "property_id": 0,
314             "property_name": "id",
315             "property_type": {
316                 "primitive_type": "dt_signed_int64"
317             }
318         },
319         {
320             "property_id": 1,
321             "property_name": "type",
322             "property_type": {
323                 "varchar": {
324                     "max_length": 64
325                 }
326             }
327         },
328         {
329             "property_id": 2,
330             "property_name": "name",
331             "property_type": {
332                 "varchar": {
333                     "max_length": 256
334                 }
335             }
336         },
337         {

```

```

338         "property_id": 3,
339         "property_name": "url",
340         "property_type": {
341             "varchar": {
342                 "max_length": 256
343             }
344         }
345     },
346 ],
347 "primary_keys": [
348     "id"
349 ]
350 },
351 {
352     "type_id": 6,
353     "type_name": "tagclass",
354     "properties": [
355         {
356             "property_id": 0,
357             "property_name": "id",
358             "property_type": {
359                 "primitive_type": "dt_signed_int64"
360             }
361         },
362         {
363             "property_id": 1,
364             "property_name": "name",
365             "property_type": {
366                 "varchar": {
367                     "max_length": 256
368                 }
369             }
370         },
371         {
372             "property_id": 2,
373             "property_name": "url",
374             "property_type": {
375                 "varchar": {
376                     "max_length": 256
377                 }
378             }
379         }
380     ],
381     "primary_keys": [
382         "id"
383     ]
384 },
385 {
386     "type_id": 7,
387     "type_name": "tag",
388     "properties": [
389         {
390             "property_id": 0,
391             "property_name": "id",
392             "property_type": {
393                 "primitive_type": "dt_signed_int64"
394             }
395         },

```

```

396         {
397             "property_id": 1,
398             "property_name": "name",
399             "property_type": {
400                 "varchar": {
401                     "max_length": 256
402                 }
403             }
404         },
405         {
406             "property_id": 2,
407             "property_name": "url",
408             "property_type": {
409                 "varchar": {
410                     "max_length": 256
411                 }
412             }
413         }
414     ],
415     "primary_keys": [
416         "id"
417     ]
418 },
419 ],
420 "edge_types": [
421     {
422         "type_id": 0,
423         "type_name": "hascreator",
424         "vertex_type_pair_relations": [
425             {
426                 "source_vertex": "comment",
427                 "destination_vertex": "person",
428                 "relation": "many_to_one",
429                 "x_csr_params": {
430                     "sort_on_compaction": true
431                 }
432             },
433             {
434                 "source_vertex": "post",
435                 "destination_vertex": "person",
436                 "relation": "many_to_one",
437                 "x_csr_params": {
438                     "sort_on_compaction": true
439                 }
440             }
441         ],
442         "properties": [
443             {
444                 "property_id": 0,
445                 "property_name": "creationdate",
446                 "property_type": {
447                     "date": "date"
448                 }
449             }
450         ]
451     },
452     {
453         "type_id": 1,

```

```

454         "type_name": "hastag",
455         "vertex_type_pair_relations": [
456             {
457                 "source_vertex": "post",
458                 "destination_vertex": "tag",
459                 "relation": "many_to_many"
460             }
461         ]
462     },
463     {
464         "type_id": 2,
465         "type_name": "replyof",
466         "vertex_type_pair_relations": [
467             {
468                 "source_vertex": "comment",
469                 "destination_vertex": "comment",
470                 "relation": "many_to_one"
471             },
472             {
473                 "source_vertex": "comment",
474                 "destination_vertex": "post",
475                 "relation": "many_to_one"
476             }
477         ]
478     },
479     {
480         "type_id": 3,
481         "type_name": "containerof",
482         "vertex_type_pair_relations": [
483             {
484                 "source_vertex": "forum",
485                 "destination_vertex": "post",
486                 "relation": "one_to_many",
487                 "x_csr_params": {
488                     "edge_storage_strategy": "only_in"
489                 }
490             }
491         ]
492     },
493     {
494         "type_id": 4,
495         "type_name": "hasmember",
496         "vertex_type_pair_relations": [
497             {
498                 "source_vertex": "forum",
499                 "destination_vertex": "person",
500                 "relation": "many_to_many",
501                 "x_csr_params": {
502                     "edge_storage_strategy": "only_in",
503                     "sort_on_compaction": true
504                 }
505             }
506         ],
507         "properties": [
508             {
509                 "property_id": 0,
510                 "property_name": "joindate",
511                 "property_type": {

```

```

512         "date": "date"
513     }
514 }
515 ]
516 },
517 {
518     "type_id": 5,
519     "type_name": "hasmoderator",
520     "vertex_type_pair_relations": [
521         {
522             "source_vertex": "forum",
523             "destination_vertex": "person",
524             "relation": "many_to_one",
525             "x_csr_params": {
526                 "edge_storage_strategy": "only_out"
527             }
528         }
529     ]
530 },
531 {
532     "type_id": 6,
533     "type_name": "hasinterest",
534     "vertex_type_pair_relations": [
535         {
536             "source_vertex": "person",
537             "destination_vertex": "tag",
538             "relation": "many_to_many",
539             "x_csr_params": {
540                 "edge_storage_strategy": "only_out"
541             }
542         }
543     ]
544 },
545 {
546     "type_id": 7,
547     "type_name": "islocatedin",
548     "vertex_type_pair_relations": [
549         {
550             "source_vertex": "comment",
551             "destination_vertex": "place",
552             "relation": "many_to_one"
553         },
554         {
555             "source_vertex": "person",
556             "destination_vertex": "place",
557             "relation": "many_to_one"
558         },
559         {
560             "source_vertex": "post",
561             "destination_vertex": "place",
562             "relation": "many_to_one"
563         },
564         {
565             "source_vertex": "organisation",
566             "destination_vertex": "place",
567             "relation": "many_to_one",
568             "x_csr_params": {
569                 "oe_mutability": "immutable"

```



```

570         }
571     }
572 ]
573 },
574 {
575     "type_id": 8,
576     "type_name": "knows",
577     "vertex_type_pair_relations": [
578         {
579             "source_vertex": "person",
580             "destination_vertex": "person",
581             "relation": "many_to_many"
582         }
583     ],
584     "properties": [
585         {
586             "property_id": 0,
587             "property_name": "creationdate",
588             "property_type": {
589                 "date": "date"
590             }
591         }
592     ]
593 },
594 {
595     "type_id": 9,
596     "type_name": "likes",
597     "vertex_type_pair_relations": [
598         {
599             "source_vertex": "person",
600             "destination_vertex": "comment",
601             "relation": "many_to_many",
602             "x_csr_params": {
603                 "edge_storage_strategy": "only_in"
604             }
605         },
606         {
607             "source_vertex": "person",
608             "destination_vertex": "post",
609             "relation": "many_to_many",
610             "x_csr_params": {
611                 "edge_storage_strategy": "only_in"
612             }
613         }
614     ],
615     "properties": [
616         {
617             "property_id": 0,
618             "property_name": "creationdate",
619             "property_type": {
620                 "date": "date"
621             }
622         }
623     ]
624 },
625 {
626     "type_id": 10,
627     "type_name": "workat",

```

```

628         "vertex_type_pair_relations": [
629             {
630                 "source_vertex": "person",
631                 "destination_vertex": "organisation",
632                 "relation": "many_to_many"
633             }
634         ],
635         "properties": [
636             {
637                 "property_id": 0,
638                 "property_name": "workfrom",
639                 "property_type": {
640                     "primitive_type": "dt_signed_int32"
641                 }
642             }
643         ]
644     },
645     {
646         "type_id": 11,
647         "type_name": "ispartof",
648         "vertex_type_pair_relations": [
649             {
650                 "source_vertex": "place",
651                 "destination_vertex": "place",
652                 "relation": "many_to_one",
653                 "x_csr_params": {
654                     "oe_mutability": "immutable"
655                 }
656             }
657         ]
658     },
659     {
660         "type_id": 12,
661         "type_name": "hastype",
662         "vertex_type_pair_relations": [
663             {
664                 "source_vertex": "tag",
665                 "destination_vertex": "tagclass",
666                 "relation": "many_to_one",
667                 "x_csr_params": {
668                     "oe_mutability": "immutable"
669                 }
670             }
671         ]
672     },
673     {
674         "type_id": 13,
675         "type_name": "issubclassof",
676         "vertex_type_pair_relations": [
677             {
678                 "source_vertex": "tagclass",
679                 "destination_vertex": "tagclass",
680                 "relation": "many_to_one",
681                 "x_csr_params": {
682                     "oe_mutability": "immutable"
683                 }
684             }
685         ]

```

```

686     },
687     {
688         "type_id": 14,
689         "type_name": "studyat",
690         "vertex_type_pair_relations": [
691             {
692                 "source_vertex": "person",
693                 "destination_vertex": "organisation",
694                 "relation": "many_to_many",
695                 "x_csr_params": {
696                     "edge_storage_strategy": "only_out"
697                 }
698             }
699         ],
700         "properties": [
701             {
702                 "property_id": 0,
703                 "property_name": "classyear",
704                 "property_type": {
705                     "primitive_type": "dt_signed_int32"
706                 }
707             }
708         ]
709     }
710 ]
711 }
712 }

```

Listing A.8: content of mapping.json describing the data mapping

```

1 {
2     "graph": "ldbc_snb",
3     "loading_config": {
4         "data_source": {
5             "scheme": "file",
6             "location": "./data/social_network-sf{10,100,300,1000}-csvcomposite-stringdateformatter"
7         },
8         "import_option": "init",
9         "format": {
10             "type": "csv",
11             "metadata": {
12                 "delimiter": "|"
13             }
14         }
15     },
16     "vertex_mappings": [
17         {
18             "type_name": "place",
19             "inputs": [
20                 "static/place_0_0.csv"
21             ]
22         },
23         {
24             "type_name": "person",
25             "inputs": [
26                 "dynamic/person_0_0.csv"
27             ]
28         },

```



```

29     {
30         "type_name": "comment",
31         "inputs": [
32             "dynamic/comment_0_0.csv"
33         ]
34     },
35     {
36         "type_name": "post",
37         "inputs": [
38             "dynamic/post_0_0.csv"
39         ]
40     },
41     {
42         "type_name": "forum",
43         "inputs": [
44             "dynamic/forum_0_0.csv"
45         ]
46     },
47     {
48         "type_name": "organisation",
49         "inputs": [
50             "static/organisation_0_0.csv"
51         ]
52     },
53     {
54         "type_name": "tagclass",
55         "inputs": [
56             "static/tagclass_0_0.csv"
57         ]
58     },
59     {
60         "type_name": "tag",
61         "inputs": [
62             "static/tag_0_0.csv"
63         ]
64     }
65 ],
66 "edge_mappings": [
67     {
68         "type_triplet": {
69             "edge": "hascreator",
70             "source_vertex": "comment",
71             "destination_vertex": "person"
72         },
73         "inputs": [
74             "dynamic/comment_hascreator_person_0_0.csv.creation_date"
75         ],
76         "source_vertex_mappings": [
77             {
78                 "column": {
79                     "index": 0,
80                     "name": "id"
81                 }
82             }
83         ],
84         "destination_vertex_mappings": [
85             {
86                 "column": {

```

```

87         "index": 1,
88         "name": "id"
89     }
90 }
91 ],
92 "column_mappings": [
93     {
94         "column": {
95             "index": 2,
96             "name": "creationdate"
97         },
98         "property": "creationdate"
99     }
100 ]
101 },
102 {
103     "type_triplet": {
104         "edge": "hascreator",
105         "source_vertex": "post",
106         "destination_vertex": "person"
107     },
108     "inputs": [
109         "dynamic/post_hascreator_person_0_0.csv.creation_date"
110     ],
111     "source_vertex_mappings": [
112         {
113             "column": {
114                 "index": 0,
115                 "name": "id"
116             }
117         }
118     ],
119     "destination_vertex_mappings": [
120         {
121             "column": {
122                 "index": 1,
123                 "name": "id"
124             }
125         }
126     ],
127     "column_mappings": [
128         {
129             "column": {
130                 "index": 2,
131                 "name": "creationdate"
132             },
133             "property": "creationdate"
134         }
135     ]
136 },
137 {
138     "type_triplet": {
139         "edge": "hastag",
140         "source_vertex": "post",
141         "destination_vertex": "tag"
142     },
143     "inputs": [
144         "dynamic/post_hastag_tag_0_0.csv"

```



```

145     ],
146     "source_vertex_mappings": [
147         {
148             "column": {
149                 "index": 0,
150                 "name": "id"
151             }
152         }
153     ],
154     "destination_vertex_mappings": [
155         {
156             "column": {
157                 "index": 1,
158                 "name": "id"
159             }
160         }
161     ]
162 },
163 {
164     "type_triplet": {
165         "edge": "replyof",
166         "source_vertex": "comment",
167         "destination_vertex": "comment"
168     },
169     "inputs": [
170         "dynamic/comment_replyof_comment_0_0.csv"
171     ],
172     "source_vertex_mappings": [
173         {
174             "column": {
175                 "index": 0,
176                 "name": "id"
177             }
178         }
179     ],
180     "destination_vertex_mappings": [
181         {
182             "column": {
183                 "index": 1,
184                 "name": "id"
185             }
186         }
187     ]
188 },
189 {
190     "type_triplet": {
191         "edge": "replyof",
192         "source_vertex": "comment",
193         "destination_vertex": "post"
194     },
195     "inputs": [
196         "dynamic/comment_replyof_post_0_0.csv"
197     ],
198     "source_vertex_mappings": [
199         {
200             "column": {
201                 "index": 0,
202                 "name": "id"

```

```

203         }
204     }
205 ],
206 "destination_vertex_mappings": [
207     {
208         "column": {
209             "index": 1,
210             "name": "id"
211         }
212     }
213 ],
214 },
215 {
216     "type_triplet": {
217         "edge": "containerof",
218         "source_vertex": "forum",
219         "destination_vertex": "post"
220     },
221     "inputs": [
222         "dynamic/forum_containerof_post_0_0.csv"
223     ],
224     "source_vertex_mappings": [
225         {
226             "column": {
227                 "index": 0,
228                 "name": "id"
229             }
230         }
231     ],
232     "destination_vertex_mappings": [
233         {
234             "column": {
235                 "index": 1,
236                 "name": "id"
237             }
238         }
239     ],
240 },
241 {
242     "type_triplet": {
243         "edge": "hasmember",
244         "source_vertex": "forum",
245         "destination_vertex": "person"
246     },
247     "inputs": [
248         "dynamic/forum_hasmember_person_0_0.csv"
249     ],
250     "source_vertex_mappings": [
251         {
252             "column": {
253                 "index": 0,
254                 "name": "id"
255             }
256         }
257     ],
258     "destination_vertex_mappings": [
259         {
260             "column": {

```

```

261         "index": 1,
262         "name": "id"
263     }
264 }
265 ]
266 },
267 {
268     "type_triplet": {
269         "edge": "hasmoderator",
270         "source_vertex": "forum",
271         "destination_vertex": "person"
272     },
273     "inputs": [
274         "dynamic/forum_hasmoderator_person_0_0.csv"
275     ],
276     "source_vertex_mappings": [
277         {
278             "column": {
279                 "index": 0,
280                 "name": "id"
281             }
282         }
283     ],
284     "destination_vertex_mappings": [
285         {
286             "column": {
287                 "index": 1,
288                 "name": "id"
289             }
290         }
291     ]
292 },
293 {
294     "type_triplet": {
295         "edge": "hasinterest",
296         "source_vertex": "person",
297         "destination_vertex": "tag"
298     },
299     "inputs": [
300         "dynamic/person_hasinterest_tag_0_0.csv"
301     ],
302     "source_vertex_mappings": [
303         {
304             "column": {
305                 "index": 0,
306                 "name": "id"
307             }
308         }
309     ],
310     "destination_vertex_mappings": [
311         {
312             "column": {
313                 "index": 1,
314                 "name": "id"
315             }
316         }
317     ]
318 },

```

```

319     {
320         "type_triplet": {
321             "edge": "islocatedin",
322             "source_vertex": "comment",
323             "destination_vertex": "place"
324         },
325         "inputs": [
326             "dynamic/comment_islocatedin_place_0_0.csv"
327         ],
328         "source_vertex_mappings": [
329             {
330                 "column": {
331                     "index": 0,
332                     "name": "id"
333                 }
334             }
335         ],
336         "destination_vertex_mappings": [
337             {
338                 "column": {
339                     "index": 1,
340                     "name": "id"
341                 }
342             }
343         ]
344     },
345     {
346         "type_triplet": {
347             "edge": "islocatedin",
348             "source_vertex": "person",
349             "destination_vertex": "place"
350         },
351         "inputs": [
352             "dynamic/person_islocatedin_place_0_0.csv"
353         ],
354         "source_vertex_mappings": [
355             {
356                 "column": {
357                     "index": 0,
358                     "name": "id"
359                 }
360             }
361         ],
362         "destination_vertex_mappings": [
363             {
364                 "column": {
365                     "index": 1,
366                     "name": "id"
367                 }
368             }
369         ]
370     },
371     {
372         "type_triplet": {
373             "edge": "islocatedin",
374             "source_vertex": "post",
375             "destination_vertex": "place"
376         },

```

```

377     "inputs": [
378         "dynamic/post_islocatedin_place_0_0.csv"
379     ],
380     "source_vertex_mappings": [
381         {
382             "column": {
383                 "index": 0,
384                 "name": "id"
385             }
386         }
387     ],
388     "destination_vertex_mappings": [
389         {
390             "column": {
391                 "index": 1,
392                 "name": "id"
393             }
394         }
395     ]
396 },
397 {
398     "type_triplet": {
399         "edge": "islocatedin",
400         "source_vertex": "organisation",
401         "destination_vertex": "place"
402     },
403     "inputs": [
404         "static/organisation_islocatedin_place_0_0.csv"
405     ],
406     "source_vertex_mappings": [
407         {
408             "column": {
409                 "index": 0,
410                 "name": "id"
411             }
412         }
413     ],
414     "destination_vertex_mappings": [
415         {
416             "column": {
417                 "index": 1,
418                 "name": "id"
419             }
420         }
421     ]
422 },
423 {
424     "type_triplet": {
425         "edge": "knows",
426         "source_vertex": "person",
427         "destination_vertex": "person"
428     },
429     "inputs": [
430         "dynamic/person_knows_person_0_0.csv"
431     ],
432     "source_vertex_mappings": [
433         {
434             "column": {

```

```

435         "index": 0,
436         "name": "id"
437     }
438 }
439 ],
440 "destination_vertex_mappings": [
441     {
442         "column": {
443             "index": 1,
444             "name": "id"
445         }
446     }
447 ],
448 },
449 {
450     "type_triplet": {
451         "edge": "likes",
452         "source_vertex": "person",
453         "destination_vertex": "comment"
454     },
455     "inputs": [
456         "dynamic/person_likes_comment_0_0.csv"
457     ],
458     "source_vertex_mappings": [
459         {
460             "column": {
461                 "index": 0,
462                 "name": "id"
463             }
464         }
465     ],
466     "destination_vertex_mappings": [
467         {
468             "column": {
469                 "index": 1,
470                 "name": "id"
471             }
472         }
473     ],
474     "column_mappings": [
475         {
476             "column": {
477                 "index": 2,
478                 "name": "creationdate"
479             },
480             "property": "creationdate"
481         }
482     ]
483 },
484 {
485     "type_triplet": {
486         "edge": "likes",
487         "source_vertex": "person",
488         "destination_vertex": "post"
489     },
490     "inputs": [
491         "dynamic/person_likes_post_0_0.csv"
492     ],

```

```

493     "source_vertex_mappings": [
494         {
495             "column": {
496                 "index": 0,
497                 "name": "id"
498             }
499         }
500     ],
501     "destination_vertex_mappings": [
502         {
503             "column": {
504                 "index": 1,
505                 "name": "id"
506             }
507         }
508     ],
509     "column_mappings": [
510         {
511             "column": {
512                 "index": 2,
513                 "name": "creationdate"
514             },
515             "property": "creationdate"
516         }
517     ]
518 },
519 {
520     "type_triplet": {
521         "edge": "workat",
522         "source_vertex": "person",
523         "destination_vertex": "organisation"
524     },
525     "inputs": [
526         "dynamic/person_workat_organisation_0_0.csv"
527     ],
528     "source_vertex_mappings": [
529         {
530             "column": {
531                 "index": 0,
532                 "name": "id"
533             }
534         }
535     ],
536     "destination_vertex_mappings": [
537         {
538             "column": {
539                 "index": 1,
540                 "name": "id"
541             }
542         }
543     ],
544     "column_mappings": [
545         {
546             "column": {
547                 "index": 2,
548                 "name": "workfrom"
549             },
550             "property": "workfrom"

```

```

551     }
552   ]
553 },
554 {
555   "type_triplet": {
556     "edge": "ispartof",
557     "source_vertex": "place",
558     "destination_vertex": "place"
559   },
560   "inputs": [
561     "static/place_ispartof_place_0_0.csv"
562   ],
563   "source_vertex_mappings": [
564     {
565       "column": {
566         "index": 0,
567         "name": "id"
568       }
569     }
570   ],
571   "destination_vertex_mappings": [
572     {
573       "column": {
574         "index": 1,
575         "name": "id"
576       }
577     }
578   ]
579 },
580 {
581   "type_triplet": {
582     "edge": "hastype",
583     "source_vertex": "tag",
584     "destination_vertex": "tagclass"
585   },
586   "inputs": [
587     "static/tag_hastype_tagclass_0_0.csv"
588   ],
589   "source_vertex_mappings": [
590     {
591       "column": {
592         "index": 0,
593         "name": "id"
594       }
595     }
596   ],
597   "destination_vertex_mappings": [
598     {
599       "column": {
600         "index": 1,
601         "name": "id"
602       }
603     }
604   ]
605 },
606 {
607   "type_triplet": {
608     "edge": "issubclassof",

```

```

609         "source_vertex": "tagclass",
610         "destination_vertex": "tagclass"
611     },
612     "inputs": [
613         "static/tagclass_issubclassof_tagclass_0_0.csv"
614     ],
615     "source_vertex_mappings": [
616         {
617             "column": {
618                 "index": 0,
619                 "name": "id"
620             }
621         }
622     ],
623     "destination_vertex_mappings": [
624         {
625             "column": {
626                 "index": 1,
627                 "name": "id"
628             }
629         }
630     ]
631 },
632 {
633     "type_triplet": {
634         "edge": "studyat",
635         "source_vertex": "person",
636         "destination_vertex": "organisation"
637     },
638     "inputs": [
639         "dynamic/person_studyat_organisation_0_0.csv"
640     ],
641     "source_vertex_mappings": [
642         {
643             "column": {
644                 "index": 0,
645                 "name": "id"
646             }
647         }
648     ],
649     "destination_vertex_mappings": [
650         {
651             "column": {
652                 "index": 1,
653                 "name": "id"
654             }
655         }
656     ],
657     "column_mappings": [
658         {
659             "column": {
660                 "index": 2,
661                 "name": "classyear"
662             },
663             "property": "classyear"
664         }
665     ]
666 }

```

```

667     ]
668 }

```

A.5 benchmark configuration

Listing A.9: contents of `interactive-benchmark-sf100.properties` used for scale factor 100

```

1
2 url=http://127.0.0.1:18089
3
4 printquerynames=false
5 printquerystrings=false
6 printqueryresults=false
7
8 status=1
9 name=ldbc-snb
10 mode=execute_benchmark
11 results_log=true
12 time_unit=microseconds
13 peer_identifiers=
14 workload_statistics=false
15 spinner_wait_duration=1
16 help=false
17 ignore_scheduled_start_times=false
18
19 workload=org.ldbcouncil.snb.driver.workloads.interactive.ldbcsnbinteractiveworkload
20 db=org.ldbcouncil.snb.impls.workloads.galaxybase.interactive.galaxybaseinteractivedb
21
22 ldbc.snb.interactive.updates_dir=/driver/data/social_network-sf100-numpart-48
23 ldbc.snb.interactive.parameters_dir=/driver/data/substitution_parameters-sf100
24 ldbc.snb.interactive.short_read_dissipation=0.2
25 ldbc.snb.interactive.scale_factor=100
26
27 thread_count=96
28 time_compression_ratio=0.001
29 operation_count=1030000000
30 warmup=241000000
31
32 ldbc.snb.interactive.ldbcquery1_enable=true
33 ldbc.snb.interactive.ldbcquery2_enable=true
34 ldbc.snb.interactive.ldbcquery3_enable=true
35 ldbc.snb.interactive.ldbcquery4_enable=true
36 ldbc.snb.interactive.ldbcquery5_enable=true
37 ldbc.snb.interactive.ldbcquery6_enable=true
38 ldbc.snb.interactive.ldbcquery7_enable=true
39 ldbc.snb.interactive.ldbcquery8_enable=true
40 ldbc.snb.interactive.ldbcquery9_enable=true
41 ldbc.snb.interactive.ldbcquery10_enable=true
42 ldbc.snb.interactive.ldbcquery11_enable=true
43 ldbc.snb.interactive.ldbcquery12_enable=true
44 ldbc.snb.interactive.ldbcquery13_enable=true
45 ldbc.snb.interactive.ldbcquery14_enable=true
46
47 ldbc.snb.interactive.ldbcshortquery1personprofile_enable=true
48 ldbc.snb.interactive.ldbcshortquery2personposts_enable=true
49 ldbc.snb.interactive.ldbcshortquery3personfriends_enable=true

```

```

50 ldbc.snb.interactive.ldbcshortquery4messagecontent_enable=true
51 ldbc.snb.interactive.ldbcshortquery5messagecreator_enable=true
52 ldbc.snb.interactive.ldbcshortquery6messageforum_enable=true
53 ldbc.snb.interactive.ldbcshortquery7messagereplies_enable=true
54
55 ldbc.snb.interactive.ldbcupdate1addperson_enable=true
56 ldbc.snb.interactive.ldbcupdate2addpostlike_enable=true
57 ldbc.snb.interactive.ldbcupdate3addcommentlike_enable=true
58 ldbc.snb.interactive.ldbcupdate4addforum_enable=true
59 ldbc.snb.interactive.ldbcupdate5addforummembership_enable=true
60 ldbc.snb.interactive.ldbcupdate6addpost_enable=true
61 ldbc.snb.interactive.ldbcupdate7addcomment_enable=true
62 ldbc.snb.interactive.ldbcupdate8addfriendship_enable=true

```

Listing A.10: contents of interactive-benchmark-sf300.properties used for scale factor 300

```

1 url=http://127.0.0.1:18089
2
3 printquerynames=false
4 printquerystrings=false
5 printqueryresults=false
6
7 status=1
8 name=ldbc-snb
9 mode=execute_benchmark
10 results_log=true
11 time_unit=microseconds
12 peer_identifiers=
13 workload_statistics=false
14 spinner_wait_duration=1
15 help=false
16 ignore_scheduled_start_times=false
17
18 workload=org.ldbcouncil.snb.driver.workloads.interactive.ldbcnsbinteractiveworkload
19 db=org.ldbcouncil.snb.impls.workloads.galaxybase.interactive.galaxybaseinteractivedb
20
21 ldbc.snb.interactive.updates_dir=/driver/data/social_network-sf300-numpart-48
22 ldbc.snb.interactive.parameters_dir=/driver/data/substitution_parameters-sf300
23 ldbc.snb.interactive.short_read_dissipation=0.2
24 ldbc.snb.interactive.scale_factor=300
25
26 thread_count=96
27 time_compression_ratio=0.00305
28 operation_count=1090000000
29 warmup=280000000
30
31 ldbc.snb.interactive.ldbcquery1_enable=true
32 ldbc.snb.interactive.ldbcquery2_enable=true
33 ldbc.snb.interactive.ldbcquery3_enable=true
34 ldbc.snb.interactive.ldbcquery4_enable=true
35 ldbc.snb.interactive.ldbcquery5_enable=true
36 ldbc.snb.interactive.ldbcquery6_enable=true
37 ldbc.snb.interactive.ldbcquery7_enable=true
38 ldbc.snb.interactive.ldbcquery8_enable=true
39 ldbc.snb.interactive.ldbcquery9_enable=true
40 ldbc.snb.interactive.ldbcquery10_enable=true
41 ldbc.snb.interactive.ldbcquery11_enable=true
42 ldbc.snb.interactive.ldbcquery12_enable=true

```



```

43 ldbc.snb.interactive.ldbcquery13_enable=true
44 ldbc.snb.interactive.ldbcquery14_enable=true
45
46 ldbc.snb.interactive.ldbcshortquery1personprofile_enable=true
47 ldbc.snb.interactive.ldbcshortquery2personposts_enable=true
48 ldbc.snb.interactive.ldbcshortquery3personfriends_enable=true
49 ldbc.snb.interactive.ldbcshortquery4messagecontent_enable=true
50 ldbc.snb.interactive.ldbcshortquery5messagecreator_enable=true
51 ldbc.snb.interactive.ldbcshortquery6messageforum_enable=true
52 ldbc.snb.interactive.ldbcshortquery7messagereplies_enable=true
53
54 ldbc.snb.interactive.ldbcupdate1addperson_enable=true
55 ldbc.snb.interactive.ldbcupdate2addpostlike_enable=true
56 ldbc.snb.interactive.ldbcupdate3addcommentlike_enable=true
57 ldbc.snb.interactive.ldbcupdate4addforum_enable=true
58 ldbc.snb.interactive.ldbcupdate5addforummembership_enable=true
59 ldbc.snb.interactive.ldbcupdate6addpost_enable=true
60 ldbc.snb.interactive.ldbcupdate7addcomment_enable=true
61 ldbc.snb.interactive.ldbcupdate8addfriendship_enable=true

```

Listing A.11: contents of interactive-benchmark-sf1000.properties used for scale factor 1000

```

1 url=http://127.0.0.1:18089
2
3 printquerynames=false
4 printquerystrings=false
5 printqueryresults=false
6
7 status=1
8 name=ldbc-snb
9 mode=execute_benchmark
10 results_log=true
11 time_unit=microseconds
12 peer_identifiers=
13 workload_statistics=false
14 spinner_wait_duration=1
15 help=false
16 ignore_scheduled_start_times=false
17
18 workload=org.ldbcouncil.snb.driver.workloads.interactive.ldbc_snb_interactiveworkload
19 db=org.ldbcouncil.snb.impls.workloads.galaxybase.interactive.galaxybaseinteractivedb
20
21 ldbc.snb.interactive.updates_dir=/driver/data/social_network-sf1000-numpart-48
22 ldbc.snb.interactive.parameters_dir=/driver/data/substitution_parameters-sf1000
23 ldbc.snb.interactive.short_read_dissipation=0.2
24 ldbc.snb.interactive.scale_factor=1000
25
26 thread_count=64
27 time_compression_ratio=0.0215
28 operation_count=985000000
29 warmup=245000000
30
31 ldbc.snb.interactive.ldbcquery1_enable=true
32 ldbc.snb.interactive.ldbcquery2_enable=true
33 ldbc.snb.interactive.ldbcquery3_enable=true
34 ldbc.snb.interactive.ldbcquery4_enable=true
35 ldbc.snb.interactive.ldbcquery5_enable=true
36 ldbc.snb.interactive.ldbcquery6_enable=true

```



```

37 ldbc.snb.interactive.ldbcquery7_enable=true
38 ldbc.snb.interactive.ldbcquery8_enable=true
39 ldbc.snb.interactive.ldbcquery9_enable=true
40 ldbc.snb.interactive.ldbcquery10_enable=true
41 ldbc.snb.interactive.ldbcquery11_enable=true
42 ldbc.snb.interactive.ldbcquery12_enable=true
43 ldbc.snb.interactive.ldbcquery13_enable=true
44 ldbc.snb.interactive.ldbcquery14_enable=true
45
46 ldbc.snb.interactive.ldbcshortquery1personprofile_enable=true
47 ldbc.snb.interactive.ldbcshortquery2personposts_enable=true
48 ldbc.snb.interactive.ldbcshortquery3personfriends_enable=true
49 ldbc.snb.interactive.ldbcshortquery4messagecontent_enable=true
50 ldbc.snb.interactive.ldbcshortquery5messagecreator_enable=true
51 ldbc.snb.interactive.ldbcshortquery6messageforum_enable=true
52 ldbc.snb.interactive.ldbcshortquery7messagereplies_enable=true
53
54 ldbc.snb.interactive.ldbcupdate1addperson_enable=true
55 ldbc.snb.interactive.ldbcupdate2addpostlike_enable=true
56 ldbc.snb.interactive.ldbcupdate3addcommentlike_enable=true
57 ldbc.snb.interactive.ldbcupdate4addforum_enable=true
58 ldbc.snb.interactive.ldbcupdate5addforummembership_enable=true
59 ldbc.snb.interactive.ldbcupdate6addpost_enable=true
60 ldbc.snb.interactive.ldbcupdate7addcomment_enable=true
61 ldbc.snb.interactive.ldbcupdate8addfriendship_enable=true

```

A.6 validation configuration

Listing A.12: the contents of validate.properties

```

1 url=http://127.0.0.1:18089
2
3 printquerynames=false
4 printquerystrings=false
5 printqueryresults=false
6
7 status=1
8 thread_count=1
9 mode=validate_database
10 name=ldbc-snb
11 results_log=false
12 time_unit=milliseconds
13 time_compression_ratio=0.001
14 peer_identifiers=
15 workload_statistics=false
16 spinner_wait_duration=0
17 help=false
18 ignore_scheduled_start_times=true
19
20 workload=org.ldbcouncil.snb.driver.workloads.interactive.ldbcnbinteractiveworkload
21 db=org.ldbcouncil.snb.impls.workloads.galaxybase.interactive.galaxybaseinteractivedb
22
23 operation_count=10000
24
25 validate_database=/driver/data/validation_params-sf10.csv
26 ldbc.snb.interactive.parameters_dir=/driver/data/substitution_parameters-sf10

```

```

27 ldbc.snb.interactive.scale_factor=10
28 ldbc.snb.interactive.short_read_dissipation=0.2
29
30 ldbc.snb.interactive.ldbcquery1_freq=1
31 ldbc.snb.interactive.ldbcquery2_freq=1
32 ldbc.snb.interactive.ldbcquery3_freq=1
33 ldbc.snb.interactive.ldbcquery4_freq=1
34 ldbc.snb.interactive.ldbcquery5_freq=1
35 ldbc.snb.interactive.ldbcquery6_freq=1
36 ldbc.snb.interactive.ldbcquery7_freq=1
37 ldbc.snb.interactive.ldbcquery8_freq=1
38 ldbc.snb.interactive.ldbcquery9_freq=1
39 ldbc.snb.interactive.ldbcquery10_freq=1
40 ldbc.snb.interactive.ldbcquery11_freq=1
41 ldbc.snb.interactive.ldbcquery12_freq=1
42 ldbc.snb.interactive.ldbcquery13_freq=1
43 ldbc.snb.interactive.ldbcquery14_freq=1
44
45 ldbc.snb.interactive.ldbcquery1_enable=true
46 ldbc.snb.interactive.ldbcquery2_enable=true
47 ldbc.snb.interactive.ldbcquery3_enable=true
48 ldbc.snb.interactive.ldbcquery4_enable=true
49 ldbc.snb.interactive.ldbcquery5_enable=true
50 ldbc.snb.interactive.ldbcquery6_enable=true
51 ldbc.snb.interactive.ldbcquery7_enable=true
52 ldbc.snb.interactive.ldbcquery8_enable=true
53 ldbc.snb.interactive.ldbcquery9_enable=true
54 ldbc.snb.interactive.ldbcquery10_enable=true
55 ldbc.snb.interactive.ldbcquery11_enable=true
56 ldbc.snb.interactive.ldbcquery12_enable=true
57 ldbc.snb.interactive.ldbcquery13_enable=true
58 ldbc.snb.interactive.ldbcquery14_enable=true
59
60 ldbc.snb.interactive.ldbcshortquery1personprofile_enable=true
61 ldbc.snb.interactive.ldbcshortquery2personposts_enable=true
62 ldbc.snb.interactive.ldbcshortquery3personfriends_enable=true
63 ldbc.snb.interactive.ldbcshortquery4messagecontent_enable=true
64 ldbc.snb.interactive.ldbcshortquery5messagecreator_enable=true
65 ldbc.snb.interactive.ldbcshortquery6messageforum_enable=true
66 ldbc.snb.interactive.ldbcshortquery7messagereplies_enable=true
67
68 ldbc.snb.interactive.ldbcupdate1addperson_enable=true
69 ldbc.snb.interactive.ldbcupdate2addpostlike_enable=true
70 ldbc.snb.interactive.ldbcupdate3addcommentlike_enable=true
71 ldbc.snb.interactive.ldbcupdate4addforum_enable=true
72 ldbc.snb.interactive.ldbcupdate5addforummembership_enable=true
73 ldbc.snb.interactive.ldbcupdate6addpost_enable=true
74 ldbc.snb.interactive.ldbcupdate7addcomment_enable=true
75 ldbc.snb.interactive.ldbcupdate8addfriendship_enable=true

```